Hochschule Fulda
University of Applied Sciences

GLOBAL DISTRIBUTED SOFTWARE DEVELOPMENT

AI5088

Master Team Project Summer 2026 - Milestone 2 - Team 2

SkillBridge

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1. Functional Requirements Prioritized

This section expands selected Milestone 1 functional requirements into more detailed Milestone 2 requirements. The original Milestone 1 reference numbers are kept so that the requirements remain traceable from M1 to M2. Requirements are grouped first by priority and then by actor, as required in the Milestone 2 instructions.

Priority 1 Core Requirements

Priority 1 requirements are the minimum core features required for the first functional version of SkillBridge. These requirements are essential for platform security, user management, tutor discovery, communication, moderation, administration, and overall platform operation. Without these features, the main purpose of the platform cannot be achieved.

A. Unregistered Users / Guest Users

Guest users are visitors who have not registered or logged in yet. They should be able to access only limited public platform information. Core platform actions such as registration approval, messaging, payments, reporting, or tutor interaction shall require an authenticated account.

6. Tutor Search

The system shall allow guest users to search publicly visible tutor profiles by subject or keyword. It shall show only approved and publicly visible tutor profiles in guest search results. It shall display limited tutor preview information such as tutor name, subject tags, short bio, and general availability. It shall require registration and login before a guest user can contact tutors, send messages or access protected tutor details.

Actor: Guest User

20. Mobile-Responsive Interface

The system shall render the public landing page correctly on desktop and mobile devices. It shall allow guest users to browse tutor previews and registration pages on mobile devices without layout issues. It shall maintain readable text, usable navigation, and responsive page structure on smaller screens.

Actor: Guest User

B. Registered Users - Students

Students are authenticated Hochschule Fulda users who use SkillBridge to search for tutors, communicate with tutors, and access tutoring-related services.

1. Fulda-Only Registration Verification

The system shall allow registration only for users with an email address ending in "hs-fulda.de". It shall reject registration attempts using non-Fulda email addresses. It shall require email verification before a student account becomes active. It shall prevent duplicate registration using the same Fulda email address.

Actor: Student

2. Role-Based Account Creation

The system shall allow users to register as either student or tutor during account creation. It shall assign system permissions based on the selected user role. It shall prevent students from accessing tutor-only or administrator-only features. It shall store the assigned user role in the user account record.

Actor: Student / System

6. Tutor Search

The system shall allow students to search tutors by subject, keyword, language, and tutoring format. It shall display tutor search results using approved tutor profiles only. It shall display tutor information such as tutor name, subject tags, availability and profile summary in search results. It shall allow students to open a tutor profile directly from the search results page.

Actor: Student

9. In-Site Messaging

The system shall allow students to send direct messages to tutors inside the platform. It shall allow students to view message conversations with tutors. It shall display message timestamps inside chat conversations. It shall prevent students from sending messages to suspended or deleted tutor accounts. Also, guest users shall not be allowed to use the chat functionality.

Actor: Student

17. User Report Feature

The system shall allow students to report suspicious, offensive, misleading, or spam content. It shall allow students to report tutor profiles, messages, or uploaded content. It shall require students to provide a report reason before submission. It shall store submitted reports for administrator review.

Actor: Student

20. Mobile-Responsive Interface

The system shall render student dashboard pages correctly on mobile devices. It shall support responsive tutor search, profile viewing and messaging interfaces. It shall maintain usable navigation and readable content on common mobile screen sizes.

Actor: Student

26. Payment Service Flow

The system shall allow students to view a demonstration payment flow before confirming a tutoring session. It shall display payment-related information such as tutor price and selected service. It shall clearly indicate that the payment feature is for demonstration purposes only. It shall not process real financial transactions.

Actor: Student

C. Registered Users - Tutors

Tutors are authenticated Hochschule Fulda users who provide tutoring services and manage tutor-related information on the platform.

2. Role-Based Account Creation

The system shall allow tutors to access tutor-specific dashboard features after registration. It shall restrict tutor accounts from accessing administrator-only functions.

Actor: Tutor

3. Tutor Profile Creation

The system shall allow tutors to create a tutor profile after account registration. It shall allow tutors to enter profile information including subjects, academic program, semester, bio, teaching style, languages and availability. It shall allow tutors to edit their tutor profile information. It shall save tutor profile changes in the database.

Actor: Tutor

5. Tutor Profile Approval Workflow

The system shall require tutor profiles to be reviewed before becoming publicly visible. It shall mark newly created tutor profiles with a pending approval status. It shall prevent unapproved tutor profiles from appearing in tutor search results. It shall notify tutors when their profile is approved or rejected.

Actor: Tutor / System

9. In-Site Messaging

The system shall allow tutors to receive and reply to student messages inside the platform. It shall display conversation history between tutors and students. It shall show timestamps for tutor messages. It shall prevent tutors from messaging suspended or deleted student accounts.

Actor: Tutor

16. Subject Tags and Skill Badges

The system shall allow tutors to add predefined subject tags to their tutor profile. It shall allow tutors to add skill badges such as “exam prep,” “lab help,” or “bilingual support.” It shall display tutor subject tags and badges on tutor profiles. It shall use tutor subject tags in tutor search results and filtering.

Actor: Tutor

20. Mobile-Responsive Interface

The system shall render tutor dashboard pages correctly on mobile devices. It shall support responsive tutor profile editing and messaging interfaces. It shall maintain usable layouts for tutor functions on smaller screens.

Actor: Tutor

21. Media Upload Validation

The system shall validate uploaded media files before storing them. It shall check uploaded files for supported file type and maximum file size limits. It shall reject unsupported or invalid file uploads. It shall display validation error messages when upload validation fails.

Actor: Tutor / System

D. Admin / Super Admin

Admins and super admins are responsible for moderation, approval workflows, platform monitoring, and overall platform management.

2. Role-Based Account Creation

The system shall allow administrator accounts to access admin-only platform functions. It shall restrict administrators from modifying protected super admin permissions unless authorised.

Actor: Admin / Super Admin

5. Tutor Profile Approval Workflow

The system shall allow administrators to review pending tutor profile submissions. It shall allow administrators to approve or reject tutor profiles. It shall store the approval status of each tutor profile. It shall prevent rejected tutor profiles from becoming publicly visible.

Actor: Admin

17. User Report Feature

The system shall allow administrators to view reported users, profiles, messages, or content. It shall display the report reason and reporting user information to administrators. It shall allow administrators to take moderation actions such as warning, hiding, suspending or deleting content. It shall allow administrators to mark reports as resolved after review.

Actor: Admin

20. Mobile-Responsive Interface

The system shall render core admin dashboard pages correctly on desktop and tablet devices. It shall maintain readable administrative tables, moderation panels, and approval interfaces on smaller screens.

Actor: Admin

21. Media Upload Validation

The system shall allow administrators to review rejected or flagged uploads when required. It shall log upload validation failures for moderation or troubleshooting purposes.

Actor: Admin / System

24. Administrative Dashboard

The system shall provide administrators with a dashboard showing pending tutor approvals. It shall display reported content and flagged users inside the admin dashboard. It shall display basic platform statistics such as active users and pending moderation items. It shall allow administrators to navigate to moderation or approval actions directly from the dashboard.

Actor: Admin

27. Account and Content Deletion

The system shall allow administrators to deactivate or delete user accounts when required by platform policy. It shall allow administrators to remove inappropriate tutor profiles, uploaded content, or messages. It shall prevent deleted or suspended accounts from accessing the platform. It shall record administrative deletion or moderation actions for audit and moderation purposes.

Actor: Admin / System

Priority 2 Desired Requirements

Priority 2 requirements are desirable features that improve usability, trust, communication, and user engagement. These features are important for making SkillBridge more complete, but they are not the minimum core features required for the first working version.

A. Unregistered Users / Guest Users

Guest users are visitors who have not registered or logged in yet. They should be able to explore a limited public version of SkillBridge so they understand the value of the platform and are encouraged to register with a Hochschule Fulda email address. However, guest users must not receive full access to booking, messaging, private tutor details, full reviews, or downloadable learning resources.

4. Media-Rich Tutor Profile Uploads

The system shall allow guest users to view limited, approved tutor profile media, such as profile images or public introduction thumbnails. It shall hide private tutor media such as CV files, certificates, full videos, audio introductions, or sample materials from guest users. It shall show a registration or login prompt when a guest user tries to access restricted tutor media.

Actor: Guest User

7. Advanced Filtering and Sorting

The system shall allow guest users to perform a limited tutor or subject search from the public landing page. It shall allow guest users to use basic filters such as subject/category and tutoring format. It shall show guest users limited tutor preview results, such as tutor first name, subject area, short profile summary, and general rating. It shall hide full tutor profile details, contact options, booking options and private resources until the guest user registers and logs in. It shall display a clear registration call-to-action when a guest user tries to access restricted tutor details or booking features.

Actor: Guest User

8. Tutor Review and Rating System

The system shall allow guest users to see an overall tutor rating or rating summary in public preview results. It shall hide full written reviews from guest users until they register and log in. It shall use visible rating summaries to help guest users understand tutor quality and encourage registration.

Actor: Guest User

15. Resource Sharing on Tutor Profiles

The system shall allow guest users to see that a tutor has shared learning resources. It shall show only limited public resource information, such as resource title, subject and resource type.

It shall prevent guest users from opening, downloading, or accessing full learning resources. It shall prompt guest users to register or log in when they try to access full learning resources.

Actor: Guest User

22. Notification System

The system shall show guest users public call-to-action prompts, such as “Register to contact this tutor” or “Log in to save this tutor.” It shall not send personal notifications to guest users because they do not have registered accounts. It shall guide guest users toward registration when they attempt to use restricted platform features.

Actor: Guest User

B. Registered Users - Students

7. Advanced Filtering and Sorting

The system shall allow students to filter tutor search results by subject, category, language, price range, rating, availability, and tutoring format. It shall allow students to sort tutor search results by relevance, lowest price, highest rating, newest tutor profile, and best availability match. It shall update the tutor result list after filters are applied. It shall clearly display active filters so students understand why certain tutors are shown.

Actor: Student

8. Tutor Review and Rating System

The system shall allow students to submit a rating for a tutor after a completed tutoring interaction or session. It shall allow students to write a short review describing their learning experience with the tutor. It shall display tutor ratings and selected reviews on tutor profiles to support trust and decision-making. It shall prevent students from submitting multiple reviews for the same completed session.

Actor: Student

12. Session Request Management

The system shall allow students to view the status of their session requests, such as pending, accepted, declined, or rescheduled. It shall allow students to cancel a pending session request before it is accepted by the tutor. It shall show students any alternative time proposed by the tutor. It shall allow students to accept or reject a proposed alternative time.

Actor: Student

15. Resource Sharing on Tutor Profiles

The system shall allow students to view study resources shared on tutor profiles. It shall display resource information such as title, description, subject, file type, and upload date. It shall allow students to open or download shared resources when access is permitted. It shall clearly show whether a resource is a note, slide, link, exercise, checklist, or other study material.

Actor: Student

18. Favourites or Shortlist

The system shall allow students to save preferred tutors to a personal favourites or shortlist page. It shall allow students to remove tutors from their shortlist. It shall allow students to access their saved tutors from their dashboard. It shall preserve the student's shortlist between login sessions.

Actor: Student

22. Notification System

The system shall notify students about new messages, contact requests, session request updates, tutor responses, review status and schedule changes. It shall show unread notifications clearly inside the student dashboard or navigation area. It shall allow students to mark notifications as read.

Actor: Student

C. Registered Users — Tutors

4. Media-Rich Tutor Profile Uploads

The system shall allow tutors to upload a profile image. The system shall allow tutors to upload supporting documents such as CV files, certificates or sample materials. It shall allow tutors to upload optional media such as short introduction videos or audio introductions. It shall show uploaded media on the tutor profile after approval, where applicable. It shall store uploaded media using the approved media storage approach and save only the related file information or URL in the database.

Actor: Tutor

8. Tutor Review and Rating System

The system shall allow tutors to view ratings and reviews submitted by students. It shall show tutors their average rating based on submitted student ratings. It shall allow tutors to report inappropriate or unfair reviews for admin review.

Actor: Tutor

12. Session Request Management

The system shall allow tutors to view incoming session requests from students. It shall allow tutors to accept a student session request. It shall allow tutors to decline a student session request. It shall allow tutors to propose a new time when the requested time is not suitable. It shall update the session request status after the tutor takes action.

Actor: Tutor

15. Resource Sharing on Tutor Profiles

The system shall allow tutors to upload study resources such as notes, slides, links, exercises, example solutions, and preparation checklists. It shall allow tutors to add a title, subject/category, description, and file type for each shared resource. It shall allow tutors to edit or delete their own shared resources. It shall mark new or edited shared resources for admin review or AI-supported moderation when required.

Actor: Tutor

22. Notification System

The system shall notify tutors about new messages, session requests, accepted or cancelled sessions, student reviews, and profile approval updates. It shall show unread tutor notifications clearly inside the tutor dashboard or navigation area. It shall allow tutors to mark notifications as read.

Actor: Tutor

23. Tutor Visibility Controls

The system shall allow tutors to temporarily hide their public tutor profile. It shall allow tutors to pause availability without deleting their profile. It shall allow tutors to mark themselves as “not accepting requests.” It shall prevent hidden or paused tutor profiles from appearing as available in normal student search results. It shall allow tutors to reactivate their profile or availability when they are ready to accept requests again.

Actor: Tutor

D. Admin / Super Admin

4. Media-Rich Tutor Profile Uploads

The system shall allow admins to review uploaded tutor media when approval is required. It shall allow admins to approve or reject uploaded media if it is inappropriate, misleading, unrelated to tutoring or violates platform rules. It shall allow admins to remove uploaded media from a tutor profile if it is reported or later found to be unsuitable.

Actor: Admin

8. Tutor Review and Rating System

The system shall allow admins to view reported reviews. It shall allow admins to hide or remove reviews that violate platform rules. It shall allow admins to keep valid reviews visible even if a tutor reports them unfairly.

Actor: Admin

15. Resource Sharing on Tutor Profiles

The system shall allow admins to review uploaded study resources when moderation is required. It shall allow admins to approve, reject, or remove tutor-uploaded resources. It shall support AI-assisted moderation to help identify inappropriate, misleading, spam-like or copyright-sensitive learning resources before they become visible to students or guest users.

Actor: Admin / System

22. Notification System

The system shall notify admins about pending profile changes, reported reviews, flagged content and uploaded media requiring review. It shall allow admins to mark administrative notifications as resolved after action is taken. It shall notify admins when AI-supported moderation flags tutor content, media or shared resources for manual review.

Actor: Admin / System

23. Tutor Visibility Controls

The system shall allow admins to view whether a tutor profile is active, hidden, paused or not accepting requests. It shall allow admins to override tutor visibility if a profile violates platform rules. It shall prevent hidden, suspended, or rejected tutor profiles from appearing in guest or student search results.

Actor: Admin / Super Admin

Priority 3 - Opportunistic Requirements

Priority 3 requirements are useful and innovative features, but they are not required for the core version of SkillBridge. These features can be implemented later if the team has enough time, skills, and resources.

A. Unregistered Users / Guest Users

13. Study-Trial Recommendation

The system may show guest users general tutor or subject recommendations based on the subject they search for. It shall not use personal student data for guest recommendations because the user is not logged in. It shall show a registration prompt if the guest user wants personalised recommendations based on faculty, semester, or study history.

Actor: Guest User

14. “Exam Rescue” Mode

The system may allow guest users to view a public explanation of the “Exam Rescue” feature. It may allow guest users to search for urgent exam-support subjects in a limited preview mode. It shall require registration and login before a guest user can contact tutors or send urgent exam help requests.

Actor: Guest User

19. Compare Tutors

The system may allow guest users to compare a limited number of tutor previews. It shall show only public comparison fields to guest users, such as subject, format, general rating, and short summary. It shall require login before guest users can compare full tutor profiles, detailed availability, reviews, resources, or contact options.

Actor: Guest User

25. Search Analytics for Improvement

The system may record anonymous guest search trends, such as searched subjects and used filters. It shall not store personal identity information for guest users. It may use anonymous guest search data to understand which subjects attract potential new users.

Actor: Guest User / System

B. Registered Users — Students

10. Tutor Contact Request

The system shall allow students to send a contact request to a tutor from the tutor profile page. It shall allow students to send a contact request from the tutor search results page. It shall allow students to include a short message explaining their tutoring need. It shall show the student whether the contact request is pending, accepted, or declined.

Actor: Student

11. Availability Calendar

The system shall allow students to view a tutor’s weekly availability calendar. It shall show available, unavailable, and already booked time windows. It shall allow students to use the availability calendar before sending a contact or booking request.

Actor: Student

13. Study-Trial Recommendation

The system shall suggest tutors based on the student's faculty, semester, selected subjects, and previous search behaviour. It shall recommend tutors who match prerequisite topics or related academic needs. It shall display recommended tutors separately from normal search results. It shall allow students to ignore recommendations and continue with manual search.

Actor: Student

14. "Exam Rescue" Mode

The system shall allow students to mark a tutoring need as urgent. It shall allow students to select an urgency reason, such as exam preparation, assignment deadline or last-minute help. It shall highlight tutors with the nearest matching availability. It shall show urgent search results separately or with a clear "Exam Rescue" label.

Actor: Student

19. Compare Tutors

The system shall allow students to select multiple tutors for comparison. It shall display selected tutors side by side. It shall compare tutors using key attributes such as subject, price, rating, language, availability, tutoring format, and media content. It shall allow students to remove tutors from the comparison view.

Actor: Student

C. Registered Users — Tutors

10. Tutor Contact Request

The system shall allow tutors to receive contact requests from students. It shall allow tutors to accept or decline contact requests. It shall allow tutors to view the student's short request message before responding.

Actor: Tutor

11. Availability Calendar

The system shall allow tutors to create and update weekly availability time windows. It shall allow tutors to remove unavailable time slots. It shall prevent already booked time slots from being shown as available.

Actor: Tutor

14. "Exam Rescue" Mode

The system shall allow tutors to decide whether they want to appear in urgent "Exam Rescue" search results. It shall allow tutors to set short-notice availability for urgent requests. It shall allow tutors to turn off urgent request visibility when they are unavailable.

Actor: Tutor

D. Admin / Super Admin

25. Search Analytics for Improvement

The system shall record aggregated search trends such as most searched subjects, most used filters, and common search terms. It shall record common unmet search requests where students search for a subject but receive few or no results. It shall show admins basic search analytics in an admin view or report. It shall use only aggregated analytics and shall not expose private student search behaviour to tutors or other students. It shall allow admins or super admins to use search analytics to improve tutor categories, subject tags, and future platform planning.

Actor: Admin / Super Admin

2. List of Main Data Items and Entities

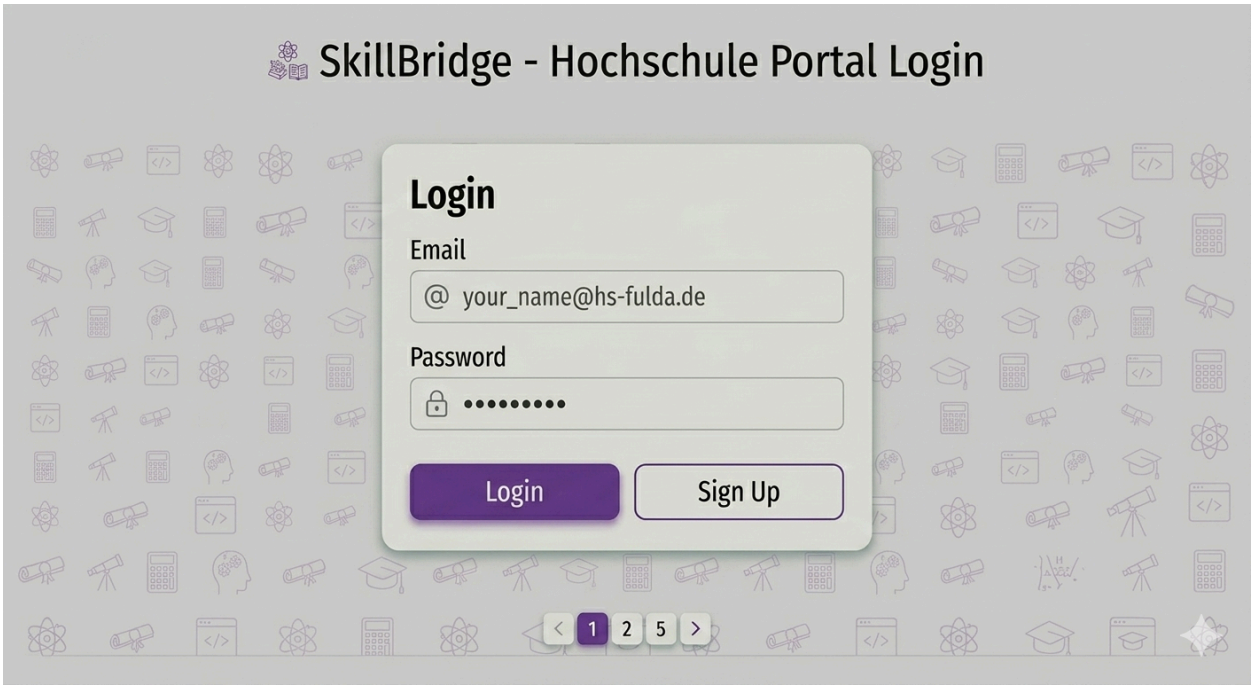
Below is a table defining key data items and entities used in the application, including columns for Index, Entity/Item Name, and Description (Purpose/Usage).

Index	Entity / Data Item	Description / Purpose
1	User	A general person using SkillBridge. A user can be a student, tutor, admin or super admin.
2	User Role	Defines the user type and privileges, such as Student, Tutor, Admin or Super Admin.
3	Student Profile	Represents student information used for finding tutors, booking sessions, viewing content and tracking activity.
4	Tutor Profile	Represents tutor information such as subjects, skills, teaching style, availability, pricing and approval status.
5	Subject / Category	Represents the academic or skill area connected to tutoring, such as programming, mathematics, language learning or exam preparation.
6	Tutor Service	Describes the type of support offered by a tutor, such as one-on-one tutoring, group tutoring or learning material support.
7	Timeslot / Availability	Represents the time periods when a tutor is available for tutoring sessions.

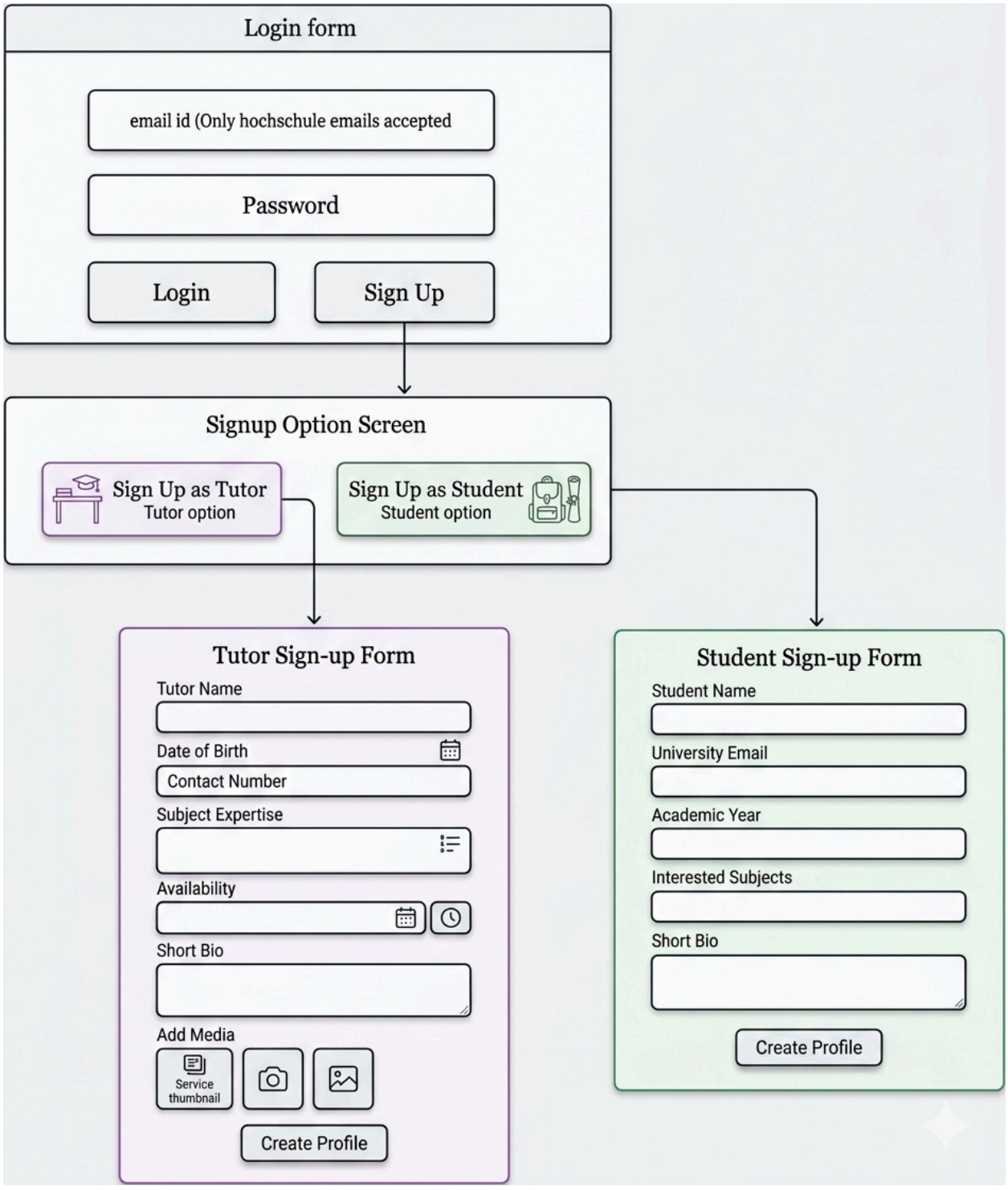
8	Booking / Session	Represents a planned tutoring appointment between a student and a tutor.
9	Learning Content	Represents study materials shared by tutors, such as notes, slides, videos, links or exercises.
10	Content Review / Moderation	Represents the approval, rejection, flagging or AI-supported review of tutor profiles, uploaded content or reported items.

3. UI Mockups

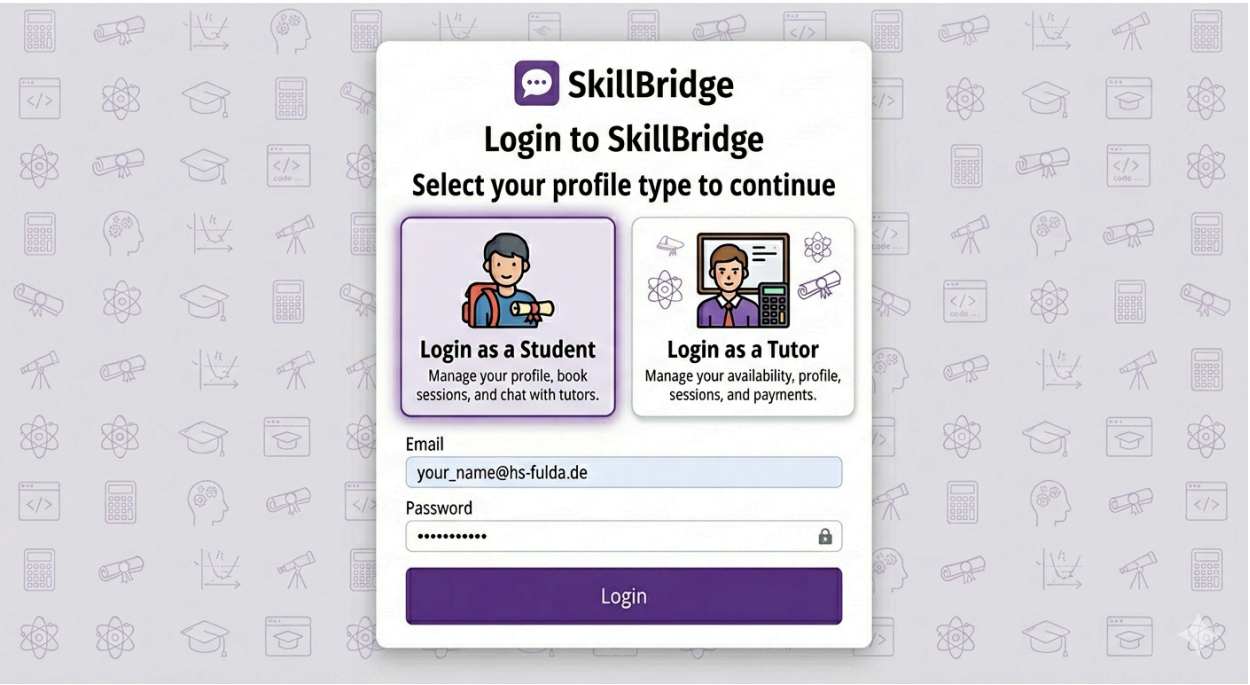
Below are the workflows and concept mockups for several Priority 1 workflows.
Our product verifies whether a student is Fulda Student or not.



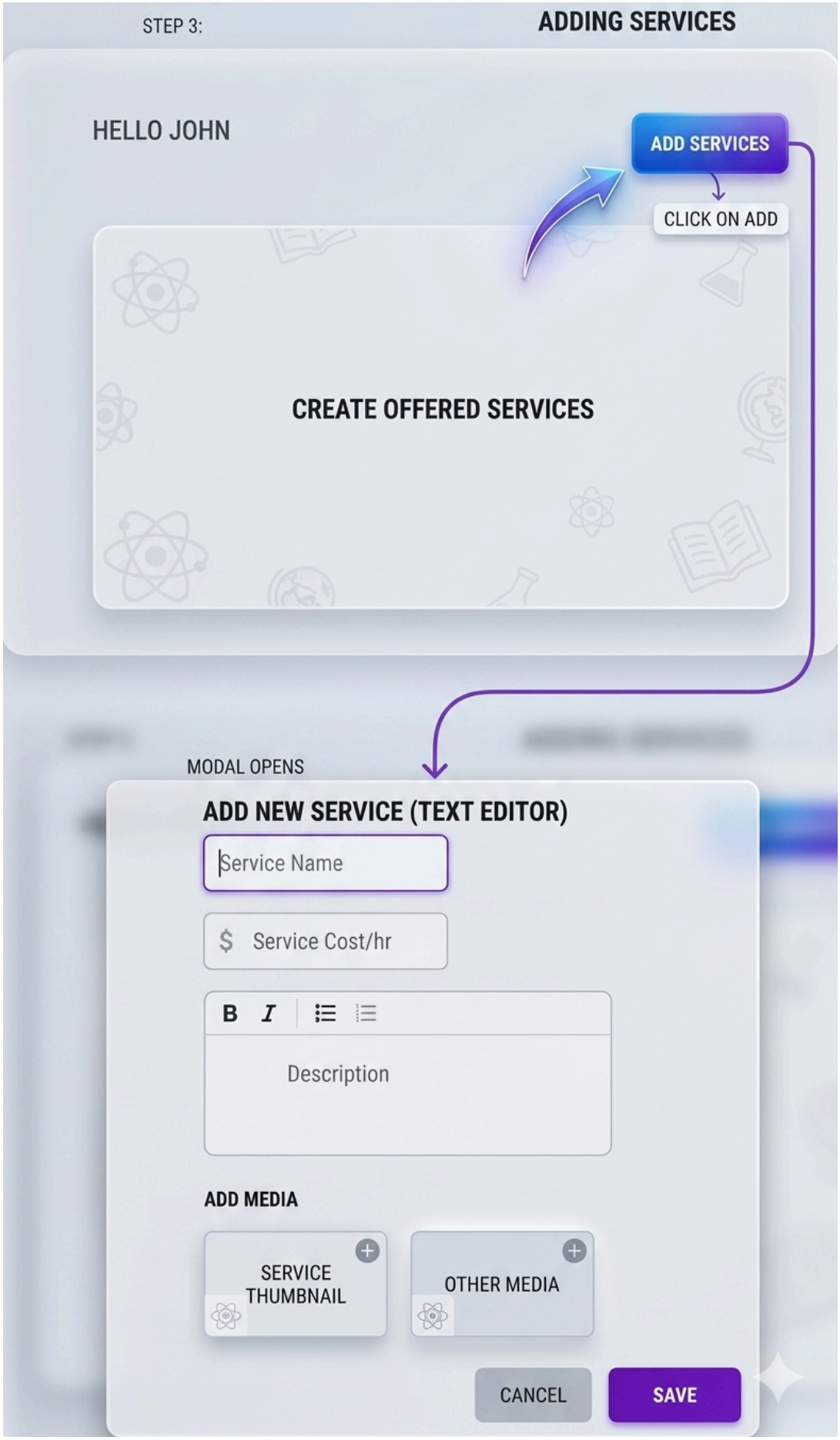
Role Based Acount creation



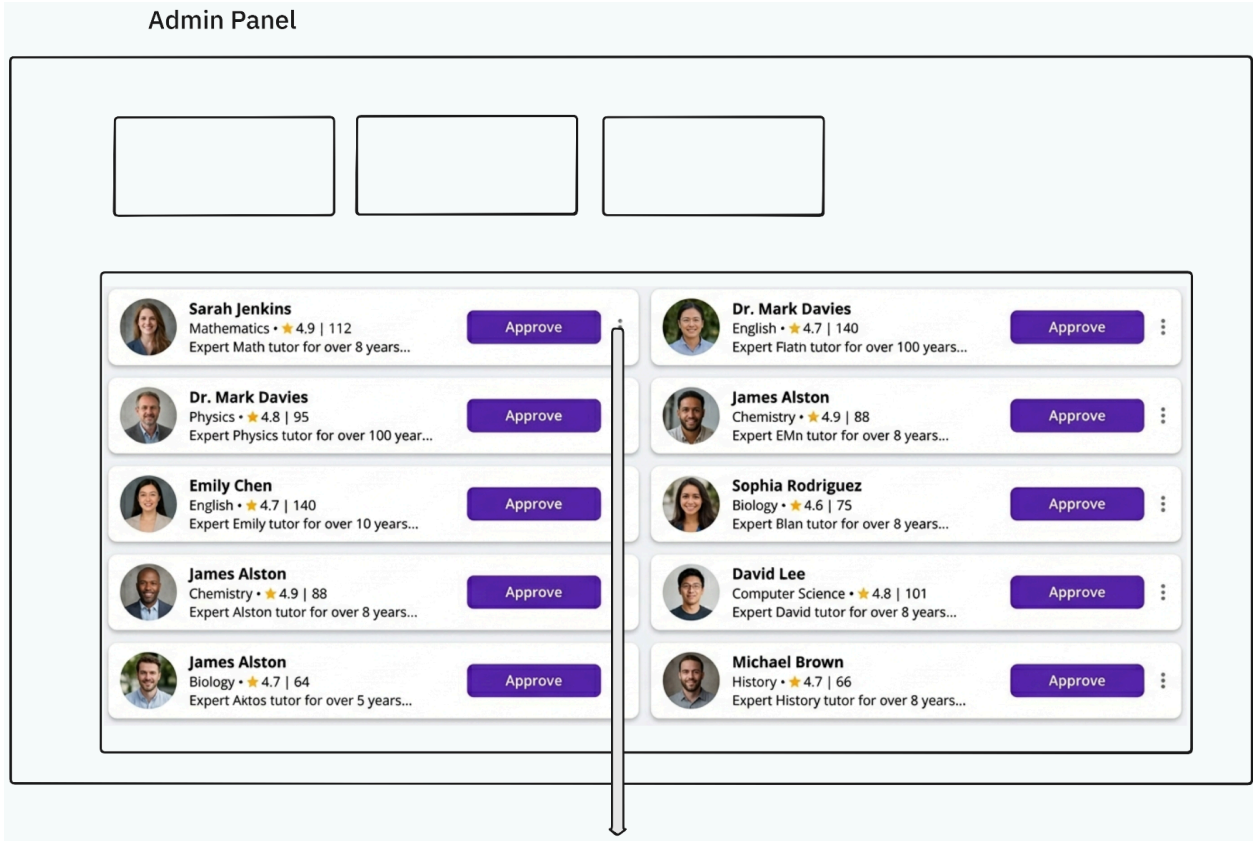
Tutor sign up and multiple service listing

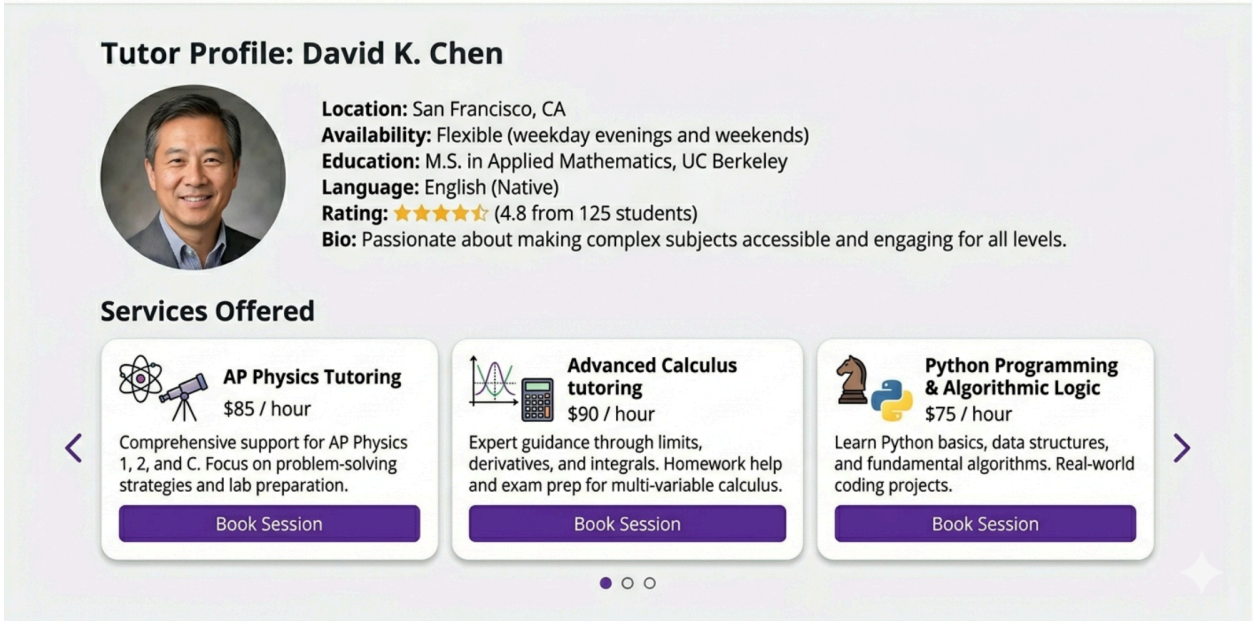


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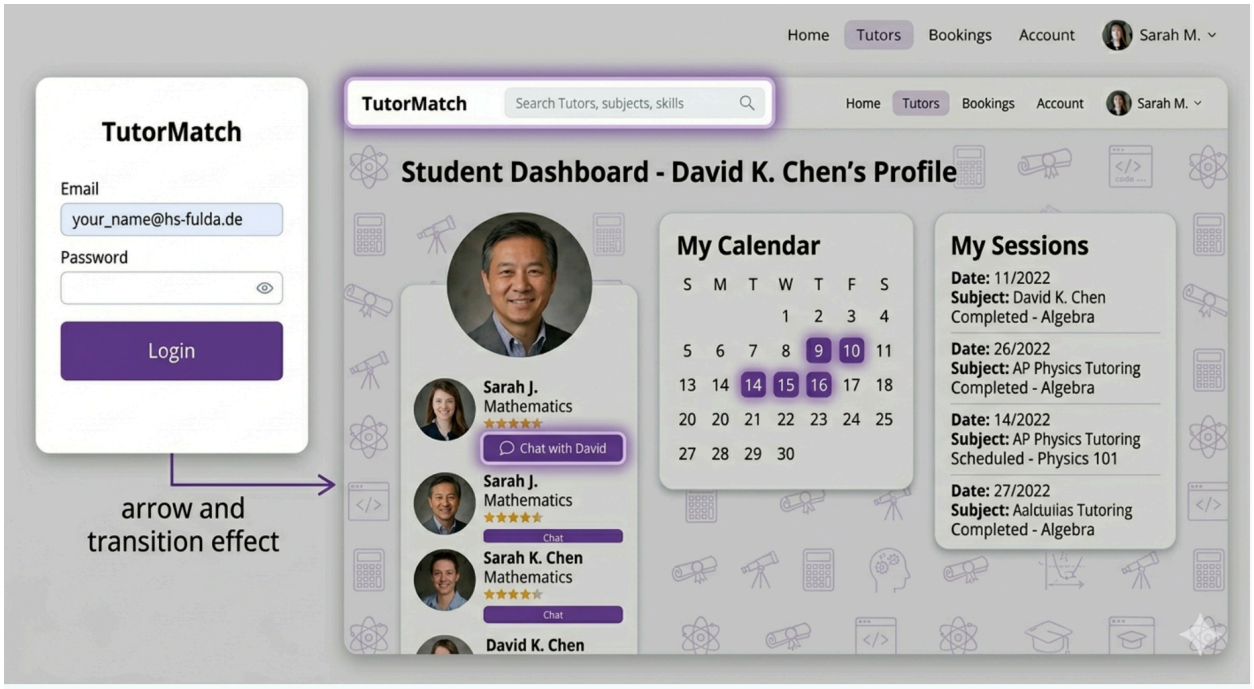

STEP 3:
ADDING SERVICES
HELLO JOHN
ADD SERVICES
CLICK ON ADD
CREATE OFFERED SERVICES
MODAL OPENS
ADD NEW SERVICE (TEXT EDITOR)
Service Name
$ Service Cost/hr
B I
Description
ADD MEDIA
SERVICE THUMBNAIL
OTHER MEDIA
CANCEL
SAVE

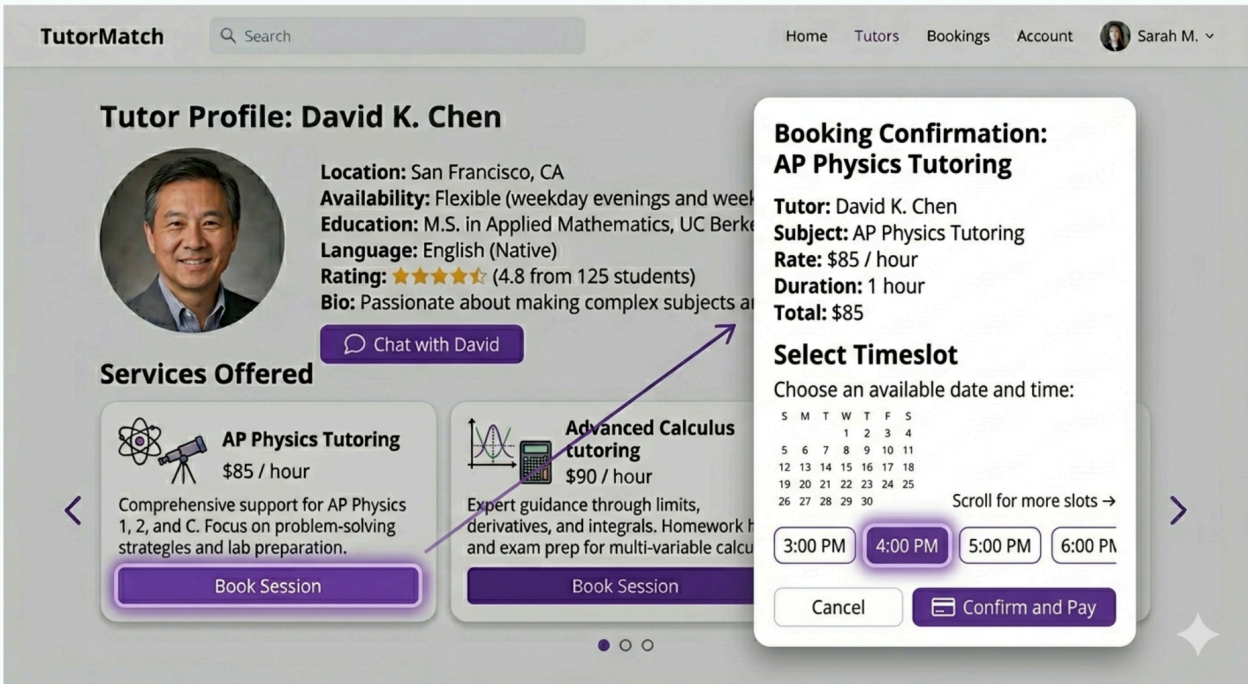
Tutor Approval Workflow

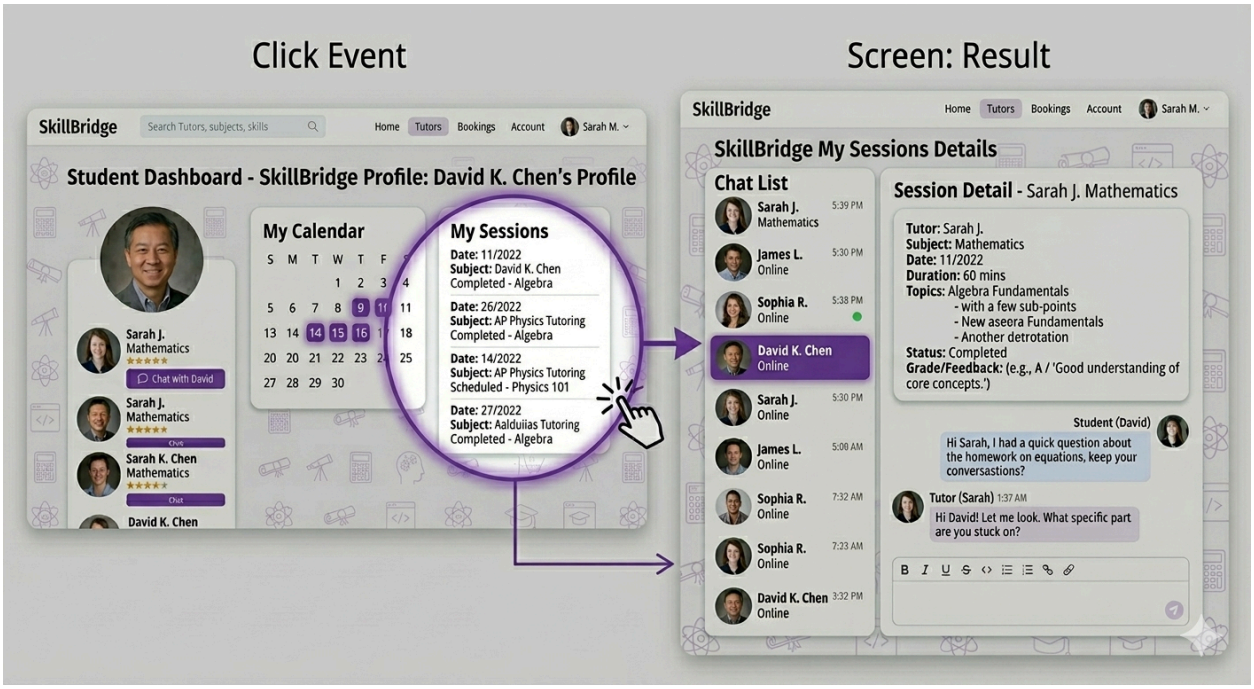




Search tutors and Chat with the tutors





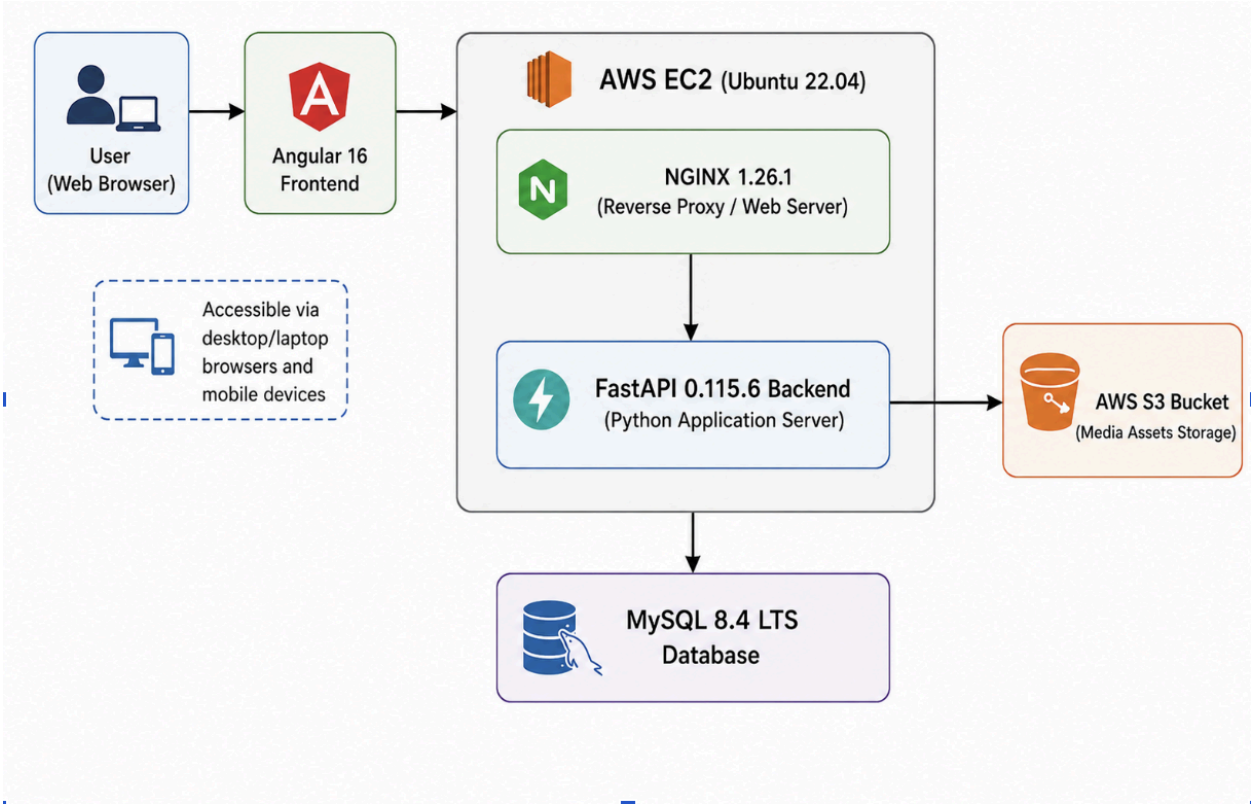


Tutor profile content moderation



4. High-Level System Architecture and Technologies Used

- **Server Host:** Amazon Web Services EC2, 1 vCPU, 2 GB RAM
- **Operating System:** Ubuntu 22.04 LTS
- **Database:** MySQL 8.4 LTS
- **Web Server:** NGINX 1.26.1
- **Server-Side Language:** Python
- **Backend Web Framework:** FastAPI 0.115.6
- **Frontend Web Framework:** Angular 16
- **IDE:** Visual Studio Code
- **Supported Browsers:** Latest two versions of Google Chrome and Mozilla Firefox
- **Deployment Platform:** AWS EC2 on Ubuntu
- **Media storage:** AWS S3 bucket for images, videos, documents, and other media assets



4.1 Media Storage Strategy

SkillBridge will use AWS S3 cloud object storage for storing uploaded images, videos, audio files, documents, and other learning materials.

The database will not store large media files directly as BLOB objects. Instead, the database will store:

- File URLs
- File metadata
- File type information
- File size
- Upload timestamps

This approach was selected because:

- It improves database performance
- It reduces database size growth
- It simplifies file scalability
- It aligns with modern cloud-based application design
- AWS S3 provides reliable media storage and retrieval

Supported media formats may include:

- Images: JPG, PNG, WEBP
- Documents: PDF, DOCX
- Video: MP4
- Audio: MP3, WAV

Media upload validation will check:

- File type
- Maximum file size
- Unsupported extensions
- Invalid uploads

4.2 Search and Filter Architecture and Implementation

SkillBridge will implement tutor search and filtering primarily using SQL-based search queries inside MySQL. The first implementation version will use:

- SQL LIKE queries
- SQL filtering conditions
- Standard relational joins
- Indexed search fields where needed

The search system will allow students to search tutors using:

- Subject
- Keywords
- Availability

- Rating
- Price range

Search-related database organisation will mainly use:
- subjects
- tutor_subjects
- tutor_profiles

Example search implementation approach:
- Subject filtering through relational joins
- Keyword matching using %LIKE%
- Sorting using SQL ORDER BY
- Pagination using SQL LIMIT

The initial project scope does not require advanced search engines such as Elasticsearch because the expected user size is relatively small for the academic project environment.

4.3 Custom APIs

SkillBridge will expose several REST APIs developed using FastAPI. These APIs will support frontend-backend communication between the Angular frontend and FastAPI backend.

Major API categories include:

1. Authentication APIs
   Purpose:
   Registration
   Login
   Email verification
   Role-based access validation

   Example endpoints:
   /auth/register
   /auth/login
   /auth/verify-email

2. Tutor Profile APIs
   Purpose:
   Tutor profile creation
   Tutor profile updates
   Tutor search
   Tutor approval workflow

   Example endpoints:
   /tutors/search
   /tutors/profile

/admin/tutors/pending

3. Messaging APIs
   Purpose:
   Conversation management
   Message sending
   Message retrieval

   Example endpoints:
   /messages/send
   /messages/conversations

4. Moderation APIs
   Purpose:
   Reporting content
   Admin review actions
   Profile approval workflows

   Example endpoints:
   /reports/create
   /admin/reports
   /admin/moderation/action

5. Session Request APIs
   Purpose:
   Tutoring session requests
   Accept/decline workflows
   Session status updates

   Example endpoints:
   /sessions/request
   /sessions/update-status

4.4 Significant Algorithms and Processes

Tutor Search Ranking

Search results may be prioritised using a simple ranking approach based on:

- Subject match
- Tutor rating
- Tutor approval status
- Availability match
- Profile completeness

The first implementation version will use SQL-based sorting and weighted ordering rather than advanced machine learning approaches.

Tutor Approval Workflow

New tutor profiles and edited profiles will follow a moderation workflow:

- Tutor submits profile
- Profile status becomes “Pending”
- Admin reviews profile
- Admin approves or rejects profile
- Only approved profiles become publicly searchable

This process improves platform quality and trust.

Report Moderation Workflow

Reported content will follow a queue-based moderation process:

- User submits report
- Report stored in moderation queue
- Admin reviews report
- Admin takes moderation action if necessary
- Report marked as resolved
- Media Validation Process

Before media files are stored:

- File type validation will occur
- File size validation will occur
- Invalid uploads will be rejected
- Valid files will be uploaded to AWS S3
- File metadata will be stored in the database

4.5 Updated Software Tools and Frameworks

The main project technology stack remains mostly unchanged from Milestone 1.
Additional libraries and frameworks may be added during implementation for:

- JWT authentication
- File upload validation
- WebSocket communication for messaging

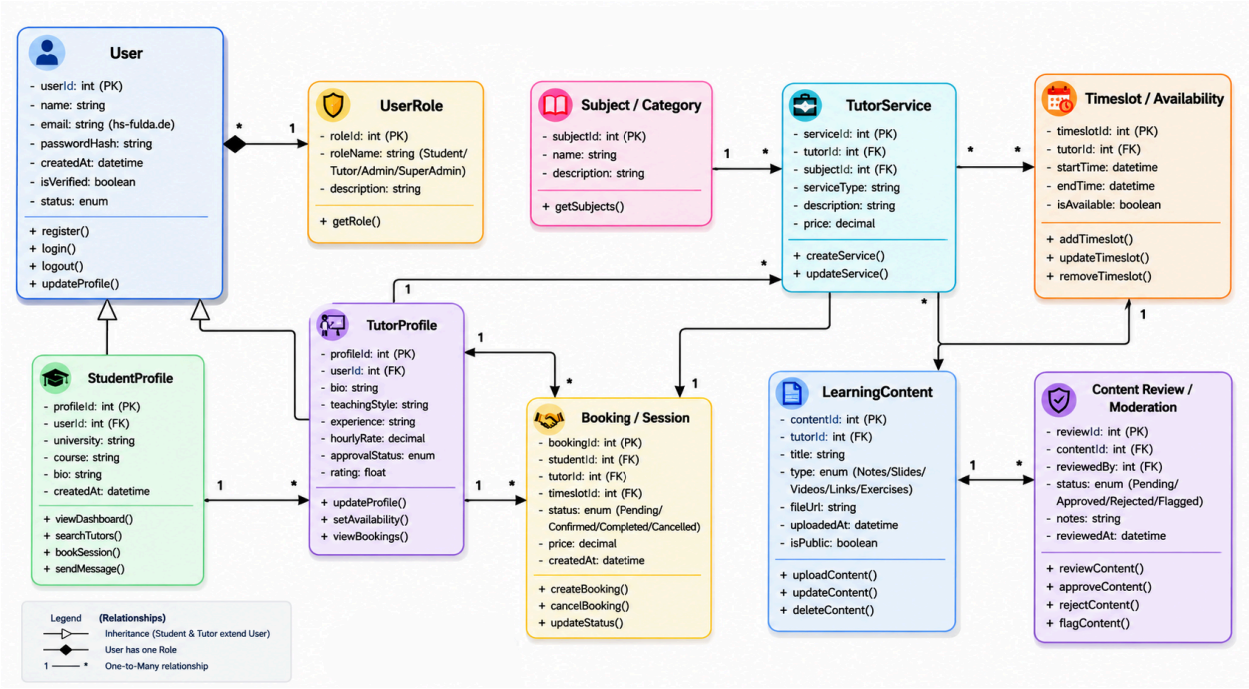
No major framework replacement is currently planned.

5. High Level UML Diagrams

The following UML diagrams represent the high-level object-oriented structure, software components and deployment architecture of the SkillBridge tutoring platform. These diagrams help visualize the relationships between core system entities, major software components and deployment infrastructure used in the project.

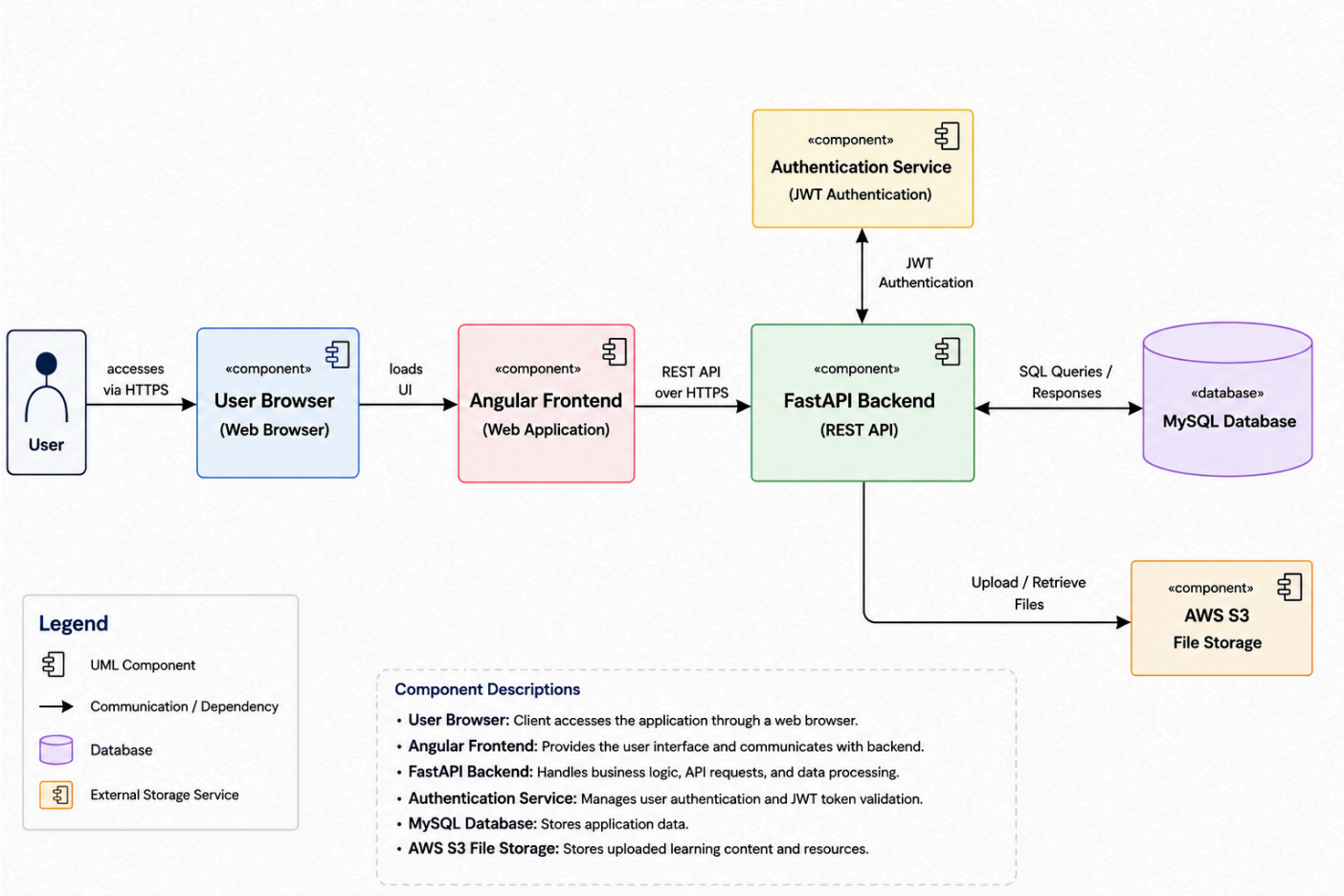
5.1 UML Class Diagram

The UML class diagram below represents the core entities, attributes and relationships used in the SkillBridge platform. It provides a high-level object-oriented view of the system structure, including users, tutors, students, booking sessions, learning content, moderation and availability management.



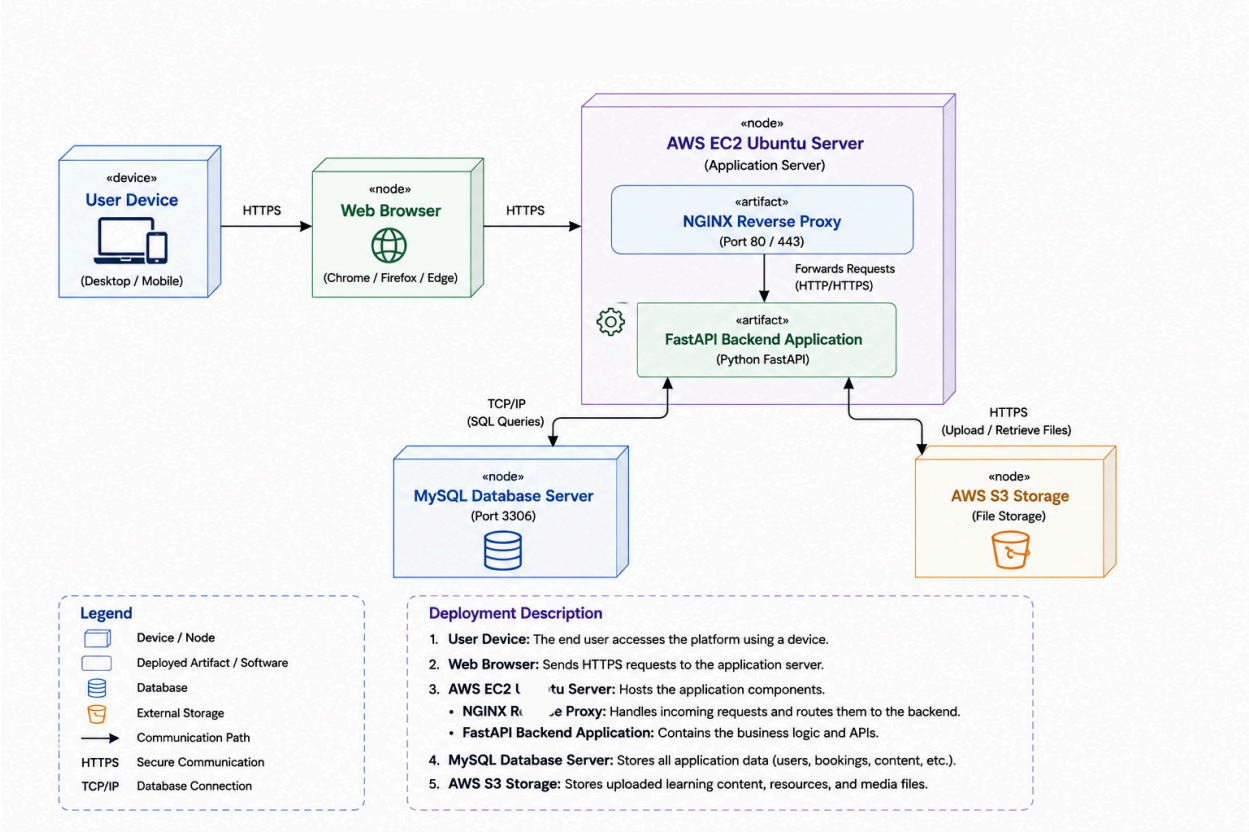
5.2 UML Component Diagram

The UML component diagram illustrates the major software components of the SkillBridge platform and the communication flow between frontend, backend, authentication services, database systems and cloud storage infrastructure.



5 .3 UML Deployment Diagram

The UML deployment diagram presents the physical deployment architecture of SkillBridge, including the client device, web browser, AWS EC2 server environment, backend application deployment, database server and cloud storage integration.


«device»
User Device
(Desktop / Mobile)
HTTPS
«node»
Web Browser
(Chrome / Firefox / Edge)
HTTPS
«node»
AWS EC2 Ubuntu Server
(Application Server)
«artifact»
NGINX Reverse Proxy
(Port 80 / 443)
Forwards Requests
(HTTP/HTTPS)
«artifact»
FastAPI Backend Application
(Python FastAPI)
TCP/IP
(SQL Queries)
HTTPS
(Upload / Retrieve Files)
«node»
MySQL Database Server
(Port 3306)
«node»
AWS S3 Storage
(File Storage)
Legend
Device / Node
Deployed Artifact / Software
Database
External Storage
Communication Path
HTTPS Secure Communication
TCP/IP Database Connection
Deployment Description
1. User Device: The end user accesses the platform using a device.
2. Web Browser: Sends HTTPS requests to the application server.
3. AWS EC2 l tu Server: Hosts the application components.
• NGINX R e Proxy: Handles incoming requests and routes them to the backend.
• FastAPI Backend Application: Contains the business logic and APIs.
4. MySQL Database Server: Stores all application data (users, bookings, content, etc.).
5. AWS S3 Storage: Stores uploaded learning content, resources, and media files.